

A six-dimensional cap for a four-torus bundle over the circle

with a machine-checked proof

Abstract

Let M^5 be the four-torus bundle over the circle whose monodromy is the unipotent matrix $T_0 = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$. We describe, and prove correct, the six-dimensional piece that is glued to the rest of a space along M^5 : it is the *solid family of four-tori over a disc*, the fibres being reglued by T_0 once round every circle of constant radius and squashed onto a two-torus over the centre. We prove that this cap is compact and connected, that its rim is a closed copy of M^5 , that every positive radius carries another closed copy of M^5 , that its centre is a closed copy of the squashed fibre, that it deformation retracts onto that centre, and — the sharpest of the statements — that the cap minus its centre is the honest collar $M^5 \times (0, 1]$, so that all of the degeneracy sits on the central two-torus. We describe the combinatorics of that central fibre (a regular hexagon with opposite sides glued: three double curves, two pinch points, every curve through both points), and we identify the closed-up model of the same data, in which nothing is collapsed, with a compact crystal $\mathbb{R}^6/\Gamma$ whose group of motions Γ is a two-step nilpotent group of affine motions — in particular that model is a nilmanifold and not a six-torus.

Every theorem stated below has been formalised and machine-checked in Lean 4 with Mathlib. The five theorems of this paper are also asserted *simultaneously*, as a single proposition `paper_certificate` in the accompanying development; it is proved without `sorry` and depends only on the three standard axioms of Lean’s logic. Section 8 is the “proof of the proof”: it explains what was formalised, how the statements in the paper correspond to the statements in the machine, and — as importantly — what is *not* claimed.

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1 Introduction

1.1 The question

Fix the four-torus $\mathbb{T}^4 = \mathbb{R}^4/\mathbb{Z}^4$ and the integral matrix

$$T_0 = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \in \mathrm{SL}_4(\mathbb{Z}), \quad N := T_0 - I, \quad N^2 = 0.$$

Since T_0 is invertible over $\mathbb{Z}$, it acts on $\mathbb{T}^4$ as a homeomorphism φ , and we may form the mapping torus

$$M^5 = (\mathbb{T}^4 \times \mathbb{R}) / ((x, t) \sim (\varphi x, t + 1)),$$

a closed five-dimensional space fibring over the circle with fibre $\mathbb{T}^4$ and monodromy φ . This is the five-manifold of the question.

The question answered here is the concrete one: *what is the six-dimensional piece with boundary M^5 that is being glued to the rest?*

1.2 The answer

The answer is a cap over a disc (Figure 1). Write the disc in polar coordinates (r, θ) , $r \in [0, 1]$. Over every point of the punctured disc put a copy of $\mathbb{T}^4$; going once round the circle of radius $r > 0$, reglue by φ . Over the centre put not a point, and not a four-torus, but the four-torus *squashed*: the two circle directions that φ moves are collapsed, and a two-torus survives. That two-torus is the hexagon with its opposite sides glued.

Two features make this the right answer rather than merely a possible one.

- The rim of the cap, $r = 1$, is exactly M^5 — that is the gluing (Theorem 4.3 (vi)).
- Away from the centre the cap is a genuine product: the cap minus its central fibre is homeomorphic to $M^5 \times (0, 1]$ (Theorem 4.3 (xiii)). So the cap is a collar on M^5 with a single degenerate fibre inserted at the centre, and the whole cap deformation retracts onto that fibre.

The squashing is not an arbitrary choice: $N^2 = 0$ forces $\mathrm{im} N \subseteq \ker N$, and for this matrix the two coincide, $\mathrm{im} N = \ker N = \langle e_0, e_1 \rangle$. Consequently $\varphi(x) - x$ always lies in the subtorus $\mathbb{T}_{\ker N}$ spanned by the first two circle directions, and the quotient $\mathbb{T}^4 \rightarrow \mathbb{T}^4/\mathbb{T}_{\ker N} \cong \mathbb{T}^2$ is the largest quotient torus on which the monodromy acts trivially. Collapsing $\mathbb{T}_{\ker N}$ is therefore the canonical way of making the fibre monodromy-invariant, which is what a fibre over the centre of the disc must be.

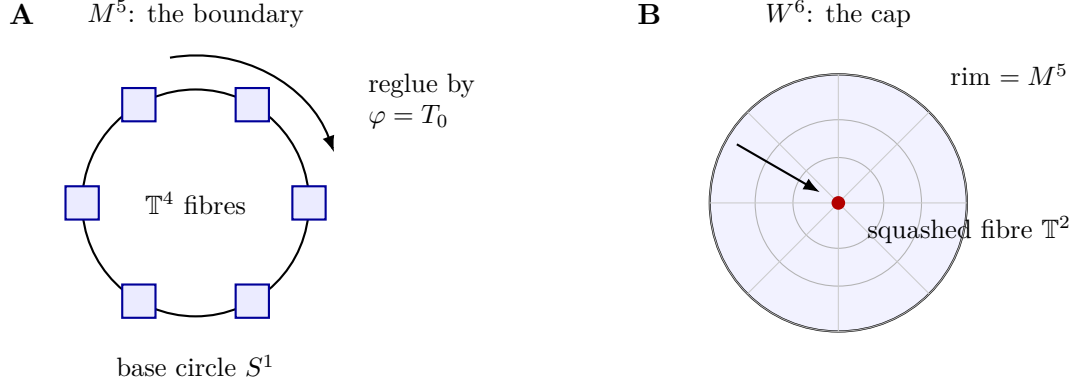


Figure 1: The boundary and the cap. (A) M^5 : four-tori over a circle, cut open and reglued by the shear $\varphi = T_0$. (B) the six-dimensional cap: the same fibres over a whole disc, with the shear round every circle of constant radius, and the squashed fibre $\mathbb{T}^2$ over the centre. The rim is M^5 ; the cap retracts onto the centre.

1.3 What is proved, and what is not

All statements in this paper are topological. M^5 and the cap are topological spaces, and we prove statements about continuous maps, closed embeddings, retractions and homeomorphisms. No smooth, symplectic or complex structure is constructed, and nothing is claimed about one. The squashing of the central fibre is an *input* to the construction — a monodromy-invariant quotient map — rather than something derived from a degeneration of complex structures; what is proved is that the resulting space has the shape drawn.

We also do not claim that the cap is a topological manifold at its centre. It is a manifold with boundary away from the centre, by the collar theorem quoted above; at the centre we prove only that a closed copy of the squashed fibre sits there and that the cap retracts onto it.

1.4 The proof of the proof

Every theorem below is accompanied by a formal counterpart in Lean 4 (Mathlib). The paper statements A–E are reproduced in the file `RequestProject/Paper.lean` as propositions `PaperA`, ..., `PaperE`, and the single theorem

`paper_certificate : PaperA ∧ PaperB ∧ PaperC ∧ PaperD ∧ PaperE`

asserts all of them at once. It is proved with no `sorry` and, as checked by `#print axioms`, depends only on `propext`, `Classical.choice` and `Quot.sound`. Section 8 gives the dictionary between the paper and the development.

1.5 Organisation

Section 2 records the monodromy data. Section 3 proves Theorem A about the boundary. Section 4 gives the general cap construction and proves Theorem B, including the collar statement. Section 5 proves Theorem C about the central fibre. Section 6 treats the closed-up model $W_\infty = \mathbb{R}^6/\Gamma$ and proves Theorem D, including three refutations of natural but false readings of the picture. Section 7 states Theorem E, the link with the cusp data the monodromy comes from. Section 8 is the formal development.

2 The monodromy

Definition 2.1. Put

$$T_1 = \begin{pmatrix} 1 & 0 & -6 & 2 \\ 0 & -1 & 1 & 1 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad T_2 = \begin{pmatrix} 1 & 6 & 0 & -3 \\ 0 & 0 & -1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad T_0 = (T_1 T_2)^{-1} = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

and $N := T_0 - I$.

Lemma 2.2. 1. T_1 has order 3 and T_2 has order 4 in $\mathrm{GL}_4(\mathbb{Z})$, and $T_1 T_2 T_0 = I$.

2. $T_0 = I + N$ with $N^2 = 0$; explicitly $Nv = (v_3, -v_2, 0, 0)$.

3. $\ker N = \mathrm{im} N = \{v : v_2 = v_3 = 0\} = \langle e_0, e_1 \rangle$.

Proof. (i) and (ii) are finite matrix computations, carried out in the formal development by `decide` and `norm_num`. For (iii): from (ii), $Nv = 0$ iff $v_2 = v_3 = 0$, and $\mathrm{im} N = \{(a, b, 0, 0)\}$ because $(v_3, -v_2)$ ranges over all of $\mathbb{Z}^2$. $\square$

Let $\varphi : \mathbb{T}^4 \rightarrow \mathbb{T}^4$ be the homeomorphism induced by T_0 ; it is a homeomorphism because T_0 has the integral inverse $T_1 T_2$. Concretely

$$\varphi(x_0, x_1, x_2, x_3) = (x_0 + x_3, x_1 - x_2, x_2, x_3). \quad (1)$$

So φ is a shear: the last two circle coordinates are untouched and are added into the first two. In particular φ fixes the subtorus $\{x_2 = x_3 = 0\}$ pointwise, which is the subtorus of $\ker N$, and $\varphi(x) - x \in \mathbb{T}_{\mathrm{im} N} = \mathbb{T}_{\ker N}$ for every x .

Lemma 2.3. $\varphi \neq \mathrm{id}$; indeed $\varphi(0, 0, 0, \frac{1}{2}) = (\frac{1}{2}, 0, 0, \frac{1}{2})$.

Proof. Immediate from (1), together with the fact that $\frac{1}{2} \notin \mathbb{Z}$, so $\frac{1}{2} \neq 0$ in $\mathbb{R}/\mathbb{Z}$. $\square$

3 The boundary five-manifold

Definition 3.1. For a homeomorphism A of a space X , the *mapping torus* is $\mathrm{MT}(A) := (X \times \mathbb{R})/\mathbb{Z}$, where $n \in \mathbb{Z}$ acts by $n \cdot (x, t) = (A^n x, t + n)$. Write $q(x, t)$ for the class of (x, t) . Put $M^5 := \mathrm{MT}(\varphi)$ and let $p : M^5 \rightarrow \mathbb{R}/\mathbb{Z}$, $p q(x, t) = t \bmod 1$.

Since the $\mathbb{Z}$ -action is by homeomorphisms, q is an open quotient map.

Theorem 3.2 (Theorem A). 1. M^5 is compact and connected.

2. p is continuous and surjective onto the circle.

3. $q(\varphi x, t + 1) = q(x, t)$ for all x, t : going once round the base applies the shear.

4. For every $a \in \mathbb{R}$ the map $\chi_a : (a, a + 1) \times \mathbb{T}^4 \rightarrow M^5$, $\chi_a(t, x) = q(x, t)$, is an open embedding, with image $p^{-1}(\text{image of } (a, a + 1) \text{ in } \mathbb{R}/\mathbb{Z})$.

In particular M^5 is a closed $\mathbb{T}^4$ -bundle over S^1 with monodromy φ .

Proof. (iii) is the definition of the identification.

(i) $M^5 = q(\mathbb{T}^4 \times [0, 1])$: given (x, t) put $n = -\lfloor t \rfloor$, so $q(x, t) = q(\varphi^n x, t + n)$ and $t + n \in [0, 1)$. Hence M^5 is a continuous image of a compact space, so compact; it is a continuous image of the connected space $\mathbb{T}^4 \times \mathbb{R}$, so connected.

(ii) p is well defined because the action shifts t by integers, continuous because q is a quotient map and $(x, t) \mapsto t \bmod 1$ is continuous, and surjective because $t \mapsto q(x_0, t)$ hits every class.

(iv) *Continuity* is clear. *Injectivity*: if $q(x, t) = q(y, s)$ with $t, s \in (a, a+1)$, then $s = t+n$ and $y = \varphi^n x$ for some $n \in \mathbb{Z}$; but $|s - t| < 1$ forces $n = 0$, whence $s = t$ and $y = x$. *Openness*: χ_a is the composite of the homeomorphism $(t, x) \mapsto (x, t)$, the open inclusion $\mathbb{T}^4 \times (a, a+1) \hookrightarrow \mathbb{T}^4 \times \mathbb{R}$ and the open map q . *Image*: the inclusion $\subseteq$ is clear. Conversely, if $p q(x, s)$ is the class of some $t \in (a, a+1)$, then $s = t + n$ for some $n \in \mathbb{Z}$, and $q(x, s) = q(\varphi^{-n} x, t) = \chi_a(t, \varphi^{-n} x)$. $\square$

4 The cap

We construct the cap in general, from three pieces of data, and then specialise.

Construction 4.1. Let X be a topological space, $A : X \rightarrow X$ a homeomorphism (the monodromy) and $c : X \rightarrow Q$ a topological quotient map with

$$c \circ A = c \quad (\text{invariance})$$

(the *collapse*). Write points of the disc in polar coordinates (r, θ) with $r \in [0, 1]$ and $\theta \in \mathbb{R}$. On $[0, 1] \times \mathbb{R} \times X$ define

$$(r, \theta, x) \sim (r', \theta', x') \iff r = r' \text{ and } \begin{cases} c(x) = c(x'), & r = 0, \\ \exists n \in \mathbb{Z} : x' = A^n x, \theta' = \theta + n, & r > 0. \end{cases}$$

Let $\text{Cap}(c) := ([0, 1] \times \mathbb{R} \times X) / \sim$, with the quotient topology, and write $[r, \theta, x]$ for classes and π for the quotient map.

Lemma 4.2. $\sim$ is an equivalence relation.

Proof. Reflexivity: if $r = 0$ take the first clause, otherwise take $n = 0$. Symmetry: the first clause is symmetric; for the second replace n by $-n$. Transitivity: the radius is constant along the relation, so the same clause applies to both steps — for $r = 0$ use transitivity of equality, for $r > 0$ add the two integers. $\square$

Define $\rho : \text{Cap}(c) \rightarrow \mathbb{R}$ by $\rho[r, \theta, x] = r$ (well defined by construction), and put

$$\iota_r : \text{MT}(A) \rightarrow \text{Cap}(c), \quad \iota_r q(x, \theta) = [r, \theta, x] \quad (r > 0), \quad \iota_0 : Q \rightarrow \text{Cap}(c), \quad \iota_0 c(x) = [0, 0, x].$$

Finally define the *radial squeeze* $H : [0, 1] \times \text{Cap}(c) \rightarrow \text{Cap}(c)$, $H_s[r, \theta, x] = [sr, \theta, x]$.

Theorem 4.3 (Theorem B). *Let X be compact, connected and nonempty. Then, with the notation above:*

- (i)–(ii) $\text{Cap}(c)$ is compact and connected;
- (iii)–(v) ρ is continuous, takes values in $[0, 1]$, and is onto $[0, 1]$;
- (vi) ι_1 is a closed embedding with image $\rho^{-1}(1)$: the rim of the cap is a closed copy of the mapping torus of the monodromy;
- (vii) more generally, for every $r > 0$, ι_r is a closed embedding with image $\rho^{-1}(r)$;
- (viii) ι_0 is a closed embedding with image $\rho^{-1}(0)$: the centre is a closed copy of the collapsed fibre Q ;
- (ix)–(xii) H is jointly continuous, $H_1 = \text{id}$, H_0 maps all of $\text{Cap}(c)$ onto the central fibre by $[r, \theta, x] \mapsto \iota_0 c(x)$, and H_s fixes the central fibre pointwise for every s ; so $\text{Cap}(c)$ deformation retracts onto its central fibre;

(xiii) the map $\Phi(m, r) := \iota_r(m)$ is a homeomorphism

$$\text{MT}(A) \times (0, 1] \xrightarrow{\cong} \{p \in \text{Cap}(c) : \rho(p) > 0\},$$

so the cap minus its central fibre is a collar on $\text{MT}(A)$.

Proof. Throughout write $\pi : [0, 1] \times \mathbb{R} \times X \rightarrow \text{Cap}(c)$ for the quotient map; a subset $S \subseteq \text{Cap}(c)$ is open (closed) precisely when $\pi^{-1}(S)$ is.

(i)–(ii). As in Theorem 3.2, $\text{Cap}(c) = \pi([0, 1] \times [0, 1] \times X)$: given (r, θ, x) put $n = -\lfloor \theta \rfloor$; if $r > 0$ then $(r, \theta, x) \sim (r, \theta + n, A^n x)$ by the second clause, and if $r = 0$ then the two are related by the first clause because $c(A^n x) = c(x)$ — invariance, iterated. Hence $\text{Cap}(c)$ is a continuous image of a compact space; and it is a continuous image of the connected space $[0, 1] \times \mathbb{R} \times X$.

(iii)–(v). ρ is well defined since the relation preserves r , continuous by the universal property of the quotient, has values in $[0, 1]$, and $\rho[r, 0, x_0] = r$ for any fixed $x_0 \in X$.

(vii) (which contains (vi)). Fix $r > 0$. *Well defined:* if $q(x, \theta) = q(y, \sigma)$, i.e. $y = A^n x$ and $\sigma = \theta + n$, then $(r, \theta, x) \sim (r, \sigma, y)$ by the second clause. *Continuity* is inherited from π through the quotient map q . *Injectivity:* if $[r, \theta, x] = [r, \sigma, y]$ then, since $r > 0$, the first clause is excluded, so $y = A^n x$ and $\sigma = \theta + n$, i.e. $q(x, \theta) = q(y, \sigma)$. *Closedness:* for any $C \subseteq \text{MT}(A)$,

$$\pi^{-1}(\iota_r(C)) = \{(r', \theta, x) : r' = r \text{ and } q(x, \theta) \in C\}. \quad (2)$$

Indeed, if $\pi(r', \theta, x) = \iota_r q(y, \sigma)$ with $q(y, \sigma) \in C$, then $r' = r$ (radii agree) and, as $r > 0$, the second clause gives $q(y, \sigma) = q(x, \theta)$, so $q(x, \theta) \in C$; the reverse inclusion is immediate. The right-hand side of (2) is the intersection of the closed set $\{r' = r\}$ with the preimage of C under the continuous map $(r', \theta, x) \mapsto q(x, \theta)$, hence closed whenever C is. A continuous, injective, closed map is a closed embedding. *Image:* clearly $\rho \iota_r = r$; conversely a point of radius r is $\pi(r, \theta, x) = \iota_r q(x, \theta)$.

(viii). ι_0 is well defined: if $c(x) = c(y)$ then $(0, 0, x) \sim (0, 0, y)$ by the first clause; and every element of Q is $c(x)$ for some x because c is onto. It is continuous because c is a quotient map and $\iota_0 \circ c = \pi(0, 0, -)$ is continuous. It is injective: if $[0, 0, x] = [0, 0, y]$ then, the radius being 0, the first clause gives $c(x) = c(y)$. It is closed by the computation

$$\pi^{-1}(\iota_0(S)) = \{(r, \theta, x) : r = 0 \text{ and } c(x) \in S\},$$

proved exactly as (2); the right-hand side is closed when S is, since c is continuous. Its image is $\rho^{-1}(0)$: a point of radius 0 is $\pi(0, \theta, x) = \iota_0 c(x)$, the angular coordinate being forgotten at the centre.

(ix)–(xii). *H is well defined.* Suppose $(r, \theta, x) \sim (r, \theta', x')$ and let $s \in [0, 1]$. If $r = 0$ then $sr = 0$ and $c(x) = c(x')$, so the images are related by the first clause. If $r > 0$ then $x' = A^n x$, $\theta' = \theta + n$; if $sr > 0$ the second clause applies verbatim, while if $sr = 0$ we need $c(A^n x) = c(x)$, which is invariance iterated. *This is exactly the point at which invariance of the collapse is used*, and it is why the fibre over the centre of a disc must be monodromy-invariant. Continuity in the two variables jointly follows because $[0, 1]$ is compact — so $\text{id} \times \pi$ is again a quotient map — and the composite $(s, (r, \theta, x)) \mapsto \pi(sr, \theta, x)$ is continuous. The identities $H_1 = \text{id}$, $H_0[r, \theta, x] = [0, \theta, x] = \iota_0 c(x)$ and $H_s \iota_0 = \iota_0$ are immediate from $1 \cdot r = r$, $0 \cdot r = 0$ and $s \cdot 0 = 0$.

(xiii). $\Phi(m, r) = \iota_r(m)$ is continuous: the map $q \times \text{id}$ from $(X \times \mathbb{R}) \times (0, 1]$ is an open continuous surjection, hence a quotient map, and the composite $((x, \theta), r) \mapsto \pi(r, \theta, x)$ is continuous. Φ is injective: $\rho \Phi(m, r) = r$ recovers r , and then ι_r is injective by (vii). Its image is $\{\rho > 0\}$, again by (vii). Finally Φ is an open map. Let $U \subseteq \text{MT}(A) \times (0, 1]$ be open. As in (2),

$$\pi^{-1}(\Phi(U)) = \{(r, \theta, x) : r > 0 \text{ and } (q(x, \theta), r) \in U\},$$

which is the preimage of U under the continuous map $(r, \theta, x) \mapsto (q(x, \theta), r)$ defined on the open set $\{r > 0\}$, hence open in $\{r > 0\}$ and therefore open in $[0, 1] \times \mathbb{R} \times X$. So $\Phi(U)$ is open, and Φ is a homeomorphism onto its image. $\square$

Construction 4.4 (the cap of the question). Take $X = \mathbb{T}^4$, $A = \varphi$ as in (1), $Q = \mathbb{T}^2$ and

$$c : \mathbb{T}^4 \rightarrow \mathbb{T}^2, \quad c(x_0, x_1, x_2, x_3) = (x_2, x_3),$$

the quotient by the subtorus $\mathbb{T}_{\ker N} = \{x_2 = x_3 = 0\}$. Then c is a quotient map (it has the continuous section $(u, v) \mapsto (0, 0, u, v)$) and $c \circ \varphi = c$ by (1). Write $W^6 := \text{Cap}(c)$.

Corollary 4.5. *W^6 is compact and connected; it has a continuous radius onto $[0, 1]$; its rim is a closed copy of M^5 ; every slice of positive radius is a closed copy of M^5 ; its centre is a closed copy of $\mathbb{T}^2$; it deformation retracts onto that centre; and*

$$\{p \in W^6 : \rho(p) > 0\} \cong M^5 \times (0, 1].$$

Proof. Theorem 4.3 applied to Construction 4.4; $\mathbb{T}^4$ is compact, connected and nonempty, and $\text{MT}(\varphi) = M^5$. $\square$

Remark 4.6. Corollary 4.5 says precisely in what sense the cap is “the solid family of four-tori over a disc”. It also isolates where it is *not* a manifold: away from the centre it is $M^5 \times (0, 1]$, a six-manifold with boundary M^5 ; at the centre a whole four-torus has been squashed onto a two-torus, and we make no claim about the local structure there.

Remark 4.7. The collapse c is canonical among quotient homomorphisms of tori: if $\mathbb{T}^4 \rightarrow T'$ is a surjective homomorphism of tori that is invariant under φ , then it kills every $\varphi(x) - x$, hence kills $\mathbb{T}_{\text{im } N} = \mathbb{T}_{\ker N}$, hence factors through c . (This remark is not part of the formalised statements.)

5 The central fibre: a hexagon with opposite sides glued

The central fibre $\mathbb{T}^2 = \mathbb{R}^2/\Lambda$ is best drawn with Λ the hexagonal lattice, whose Dirichlet domain is a regular hexagon with opposite sides glued by translations. Index the corners by $i \in \mathbb{Z}/6$: in lattice coordinates $v_i = (1, 0), (0, 1), (-1, 1), (-1, 0), (0, -1), (1, -1)$ for $i = 0, \dots, 5$, placed in the plane by $(a, b) \mapsto (a + \frac{b}{2}, \frac{\sqrt{3}}{2}b)$, so that $|v_i|^2 = a^2 + ab + b^2 = 1$. Side i runs from v_i to v_{i+1} , and the gluing vector of side i is

$$\tau_i := -(v_i + v_{i+1}).$$

Theorem 5.1 (Theorem C). *1. The drawn hexagon is regular ($|v_i| = 1$ and $|v_{i+1} - v_i| = 1$ for all i) and centrally symmetric ($v_{i+3} = -v_i$).*

2. τ_i carries side i onto side $i + 3$, taking $v_i \mapsto v_{i+4}$ and $v_{i+1} \mapsto v_{i+3}$.

3. The sides fall into exactly three classes under the gluing, $i \sim i + 3$; the three classes correspond bijectively to the three double curves of the quotient.

4. The corners fall into exactly two classes, $i \sim i + 2$, i.e. by the parity of i ; the two classes correspond bijectively to the two pinch points of the quotient.

5. Every double curve passes through both pinch points.

6. Opposite sides carry the same arrowhead: glued sides are drawn alike.

Proof. (i) With the hexagonal quadratic form $\mathbf{q}(a, b) = a^2 + ab + b^2$ one checks $\mathbf{q}(v_i) = 1$ and $\mathbf{q}(v_{i+1} - v_i) = 1$ for the six values of i , and the placement above turns $\mathbf{q}$ into the Euclidean squared length. Central symmetry $v_{i+3} = -v_i$ is read off the list. (All six checks are finite and are done by **decide** in the formalisation.)

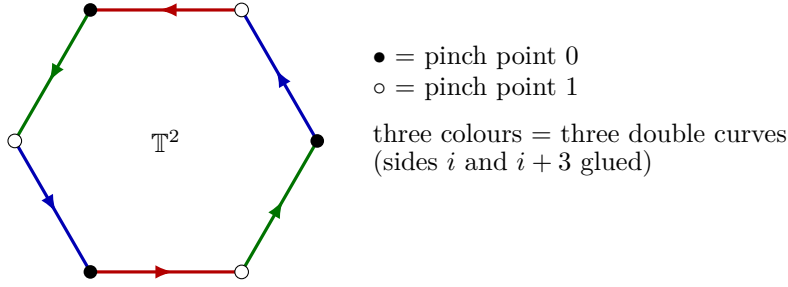


Figure 2: The squashed central fibre, drawn as a regular hexagon with opposite sides glued by the translations $\tau_i = -(v_i + v_{i+1})$. The six sides fall into three classes (the three double curves), the six corners into two (the two pinch points), and each double curve runs through both pinch points.

(ii) $v_i + \tau_i = v_i - (v_i + v_{i+1}) = -v_{i+1} = v_{i+4}$ and $v_{i+1} + \tau_i = -v_i = v_{i+3}$, using (i). So the segment $[v_i, v_{i+1}]$ is carried onto $[v_{i+4}, v_{i+3}]$, which is side $i + 3$ traversed backwards.

(iii) The identification on sides is generated by $i \sim i + 3$; the orbits are $\{0, 3\}, \{1, 4\}, \{2, 5\}$, three in number, and the reduction $\mathbb{Z}/6 \rightarrow \mathbb{Z}/3$ is a bijection on orbits.

(iv) By (ii) the gluings identify corners $v_i \sim v_{i+4}$ (and $v_{i+1} \sim v_{i+3}$). The subgroup of $\mathbb{Z}/6$ generated by 4 is $\{0, 2, 4\}$, so the orbits are the even and the odd corners: two classes, distinguished by the reduction $\mathbb{Z}/6 \rightarrow \mathbb{Z}/2$.

(v) The closure of the image of side i contains the images of the corners v_i and v_{i+1} , whose indices have opposite parity; by (iv) these are the two distinct classes. So each of the three curves meets both points.

(vi) The arrowhead on side i is drawn along the boundary for $i < 3$ and against it for $i \geq 3$, which is exactly the rule making the arrowheads on i and $i + 3$ agree as vectors, by (i). $\square$

6 Closing up: the crystal

The cap has a boundary, and a degenerate fibre in the middle. There is a second, closed object attached to the same data — the one obtained by gluing the two ends of the bar in the diagram of the question, with nothing collapsed — and it is worth recording what it is, because three natural readings of the diagram are false and this is where they are refuted.

Definition 6.1. Let $F := \mathbb{T}_h^2 \times \mathbb{T}_y^2 \times S_z^1$ (the hexagon torus, the extra $\mathbb{T}^2$ and one more circle) and let

$$g : F \rightarrow F, \quad g((h_0, h_1), (y_0, y_1), z) = ((h_0 + y_1, h_1 - y_0), (y_0, y_1), z)$$

be the same shear as φ , extended by the identity on z . Put $W_\infty := \text{MT}(g)$: the prism (hexagon torus) $\times [0, 1] \times \mathbb{T}^2 \times S^1$ with its two ends glued by g .

Definition 6.2. Let Γ act on $\mathbb{R}^6$, with coordinates $(h_0, h_1, y_0, y_1, z, t)$, by the affine motions

$$\gamma_{n,v}(h_0, h_1, y_0, y_1, z, t) = (h_0 + ny_1 + v_0, h_1 - ny_0 + v_1, y_0 + v_2, y_1 + v_3, z + v_4, t + n),$$

for $n \in \mathbb{Z}$ and $v \in \mathbb{Z}^5$.

Lemma 6.3. The maps $\gamma_{n,v}$ form a group under composition, with

$$\gamma_{n,v}\gamma_{m,w} = \gamma_{n+m,u}, \quad u = (v_0 + w_0 + nw_3, v_1 + w_1 - nv_2, v_2 + w_2, v_3 + w_3, v_4 + w_4).$$

Proof. Direct substitution: applying $\gamma_{m,w}$ and then $\gamma_{n,v}$ sends h_0 to $h_0 + my_1 + w_0 + n(y_1 + w_3) + v_0 = h_0 + (n+m)y_1 + (v_0 + w_0 + nw_3)$, and similarly for h_1 ; the other four coordinates are translated, and t is raised by $n + m$. Associativity and inverses follow from the displayed formula (the inverse of $\gamma_{n,v}$ is $\gamma_{-n,(-v_0+nv_3, -v_1-nv_2, -v_2, -v_3, -v_4)}$). $\square$

Theorem 6.4 (Theorem D). 1. Γ acts freely on $\mathbb{R}^6$.

2. Two points of $\mathbb{R}^6$ have the same image in W_∞ (under reduction of the five circle coordinates mod 1 at height t) if and only if they lie in the same Γ -orbit; consequently $\mathbb{R}^6/\Gamma \cong W_\infty$ as topological spaces.
3. The half-open unit cell $[0, 1)^6$ is a strict fundamental domain: every orbit meets it, and meets it exactly once. So copies of the cell tile $\mathbb{R}^6$.
4. Every point of W_∞ has a neighbourhood homeomorphic to an open box of $\mathbb{R}^6$; hence W_∞ is a compact connected topological six-manifold.
5. (Refutation 1.) Γ is not abelian — indeed $[\gamma_{1,0}, \gamma_{0,e_3}] = \gamma_{0,e_0} \neq 1$ — so Γ is isomorphic to no commutative group, in particular not to $\mathbb{Z}^6$: W_∞ is not a flat six-torus. Every commutator is central, so Γ is two-step nilpotent and W_∞ is a nilmanifold.
6. (Refutation 2.) The end gluing g is not a product: there are no maps f, u, k of the three drawn factors with $g(p_1, p_2, p_3) = (f(p_1), u(p_2), k(p_3))$. So the arrows along the bar cannot be read off the bar alone.
7. (Refutation 3.) No nonzero power of g is the identity: no number of trips along the bar undoes the twist.

Proof. (i) If $\gamma_{n,v}$ fixes a point then comparing t -coordinates gives $n = 0$; with $n = 0$ the motion is the translation by v , so $v = 0$.

(ii) Reduction mod 1 in the five circle coordinates, together with the mapping-torus identification in t , identifies exactly the points differing by the generators: the five unit translations, and the motion $\gamma_{1,0}$, which raises t by one and shears (h_0, h_1) by $(y_1, -y_0)$ — this is the gluing g . Since the reduction map is an open continuous surjection whose fibres are exactly the orbits, it induces a homeomorphism $\mathbb{R}^6/\Gamma \cong W_\infty$.

(iii) *Existence:* given x , first apply $\gamma_{n,0}$ with $n = -[t]$ to put t in $[0, 1)$, then translate each of the five circle coordinates by minus its floor; the composite is again an element of Γ by Lemma 6.3. *Uniqueness:* if x and $\gamma_{n,v}x$ both lie in $[0, 1)^6$, comparing t gives $|n| < 1$, so $n = 0$; then $\gamma_{0,v}$ is a translation by an integer vector preserving $[0, 1)^5$ in the remaining coordinates, so $v = 0$.

(iv) Take the open box of side $\frac{1}{2}$ centred at a point x . By the same floor argument the reduction map is injective on it, and it is open, so it carries the box homeomorphically onto an open neighbourhood of the image of x . Compactness and connectedness of W_∞ hold because it is a mapping torus of a homeomorphism of the compact connected F (Theorem 3.2(i) applies verbatim).

(v) By Lemma 6.3, $\gamma_{1,0}\gamma_{0,e_3} = \gamma_{1,e_0+e_3}$ while $\gamma_{0,e_3}\gamma_{1,0} = \gamma_{1,e_3}$; these differ, and their quotient is the unit translation γ_{0,e_0} of the first hexagon direction. An isomorphism onto a commutative group would force these two elements to commute, a contradiction. Finally the displayed product rule shows that a commutator has $n = 0$ and translation part in $\langle e_0, e_1 \rangle$, and translations in those two directions commute with everything; so every commutator is central.

(vi) Suppose $g(p_1, p_2, p_3) = (f(p_1), u(p_2), k(p_3))$ for all p . Comparing the first hexagon coordinate at $p = ((0, 0), (0, 0), 0)$ and at $p = ((0, 0), (0, \frac{1}{2}), 0)$ gives $f((0, 0)) = 0$ in the first case and $f((0, 0)) = \frac{1}{2}$ in the second, so $\frac{1}{2} = 0$ in $\mathbb{R}/\mathbb{Z}$: false.

(vii) $g^n((h_0, h_1), (y_0, y_1), z) = ((h_0 + ny_1, h_1 - ny_0), (y_0, y_1), z)$ by induction. For $n \neq 0$ take $y_1 = \frac{1}{2n}$: the first coordinate is shifted by $\frac{1}{2} \neq 0$. $\square$

Remark 6.5. W_∞ carries a continuous surjection onto M^5 commuting with the projections to the base circle (this is proved in the development), and the extra circle direction is the z -factor of Definition 6.1. The evident product decomposition $W_\infty \cong M^5 \times S^1$ is not part of

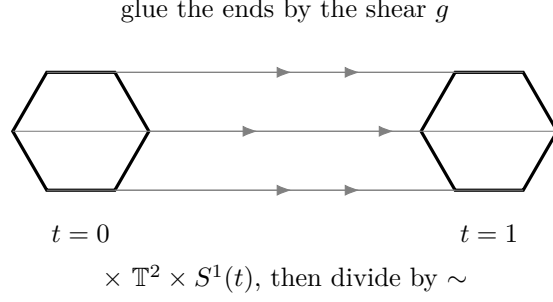


Figure 3: The closed-up model W_∞ : a bar with a hexagon at each end, the hexagon sides glued in opposite pairs, the two ends glued to each other, all of it multiplied by $\mathbb{T}^2 \times S^1$. Unfolding every identification at once presents it as the crystal $\mathbb{R}^6/\Gamma$ of Theorem 6.4.

the formalised statements. Similarly, since $\mathbb{R}^6$ is contractible and Γ acts freely and properly discontinuously, $\pi_1(W_\infty) \cong \Gamma$ and W_∞ is aspherical; the formal development proves the freeness, the fundamental domain and the charts, but not this consequence.

7 The link with the cusp data

The monodromy of Definition 2.1 does not come from nowhere: T_1 and T_2 are monodromy matrices of orders three and four, and $T_0 = (T_1 T_2)^{-1}$ is the unipotent monodromy at the corresponding cusp. The following theorem records that the object constructed above is built from exactly that data.

Theorem 7.1 (Theorem E). *$T_0 = I + N$ with $N^2 = 0$ and $T_1 T_2 T_0 = I$, where T_1 has order three and T_2 has order four. The monodromy of M^5 is the linear action of T_0 on $\mathbb{T}^4$; the circle directions it leaves fixed are those of $\ker N$; the fundamental group of M^5 is modelled by the semidirect product $\mathbb{Z}^4 \rtimes_{T_0} \mathbb{Z}$, in which conjugation by the base generator acts as T_0 ; and the combinatorics of the central fibre is the hexagon model of Theorem 5.1: three double curves, two pinch points, and every curve through both points.*

Proof. The matrix identities are Lemma 2.2. That the monodromy is the linear action of T_0 is the definition of φ , and the fixed directions were computed in (1) and Lemma 2.2(iii). The semidirect-product presentation is the standard presentation of the fundamental group of a mapping torus: in $\mathbb{Z}^4 \rtimes_{T_0} \mathbb{Z}$ one has $\iota(T_0 v) = \sigma \iota(v) \sigma^{-1}$, where ι is the inclusion of the fibre lattice and σ the base generator. The last clause is Theorem 5.1(iii)–(v). $\square$

8 The formal development: a proof of the proof

8.1 What was formalised

The development is a Lean 4 project depending on Mathlib. The correspondence between the statements above and the machine-checked statements is as follows.

paper	Lean proposition	file
Theorem 3.2	PaperA, proved by <code>panelA_valid</code>	<code>RequestProject/TorusMonodromy.lean</code>
Theorem 4.3	PaperB, proved by <code>picture_valid</code> and <code>cuspidal_cap_collar</code>	<code>RequestProject/CuspCap.lean</code> , <code>RequestProject/CuspCapT4.lean</code> , <code>RequestProject/CapPunctured.lean</code>
Theorem 5.1	PaperC, proved by <code>hexagon_picture_valid</code>	<code>RequestProject/PictureHexagon.lean</code>
Theorem 6.4	PaperD, proved by <code>crystal_picture_valid</code> , <code>cry_no_iso_commGroup</code> , <code>glueMap_zpow_ne_id</code>	<code>RequestProject/Crystal/Lattice.lean</code>
Theorem 7.1	PaperE, proved by <code>attached_project_link</code>	<code>RequestProject/AttachedProjectLink.lean</code>

The five propositions are collected in `RequestProject/Paper.lean`, where

```
theorem paper_certificate : PaperA ∧ PaperB ∧ PaperC ∧ PaperD ∧ PaperE
```

is proved. The whole project builds with `lake build` and contains no `sorry`; `#print axioms S6Family.Paper.paper_certificate` reports `[propext, Classical.choice, Quot.sound]`, the three standard axioms of Lean’s logic, so no additional assumption has been smuggled in.

8.2 Definitions in the machine

Three definitions carry the content.

- `S6Family.Picture.FiveManifold` is M^5 : the mapping torus of the homeomorphism of $(\mathbb{R}/\mathbb{Z})^4$ induced by T_0 .
- `S6Family.Picture.Cap` is Construction 4.1, taking as input a `Collapse`: a quotient map c together with a proof of $c \circ A = c$. The cap of the question is `S6Family.Picture.CuspCapT4`, the instance of Construction 4.4.
- `S6Family.Crystal.Cry` is the group Γ of Definition 6.2, defined by its six integer parameters with the multiplication of Lemma 6.3; `S6Family.Glued.WInfty` is W_∞ .

The formalisation of Theorem 4.3(xiii), the collar statement, produces an explicit homeomorphism `capPunctHomeo`, not merely an abstract existence claim; the specialised form is `cuspidal_cap_collar`.

8.3 The figures are generated from the same data

The SVG figures shipped with the project (`cuspidal-cap-6manifold-lean.svg`, `figures/crystal-diagram.svg` and the high-dimensional sheets in `figures/`) are written by Lean programs in the same development, from the very data the theorems speak about — the hexagon corners, the arrow kinds, the corner classes, the twelve ambient coordinates. Two theorems in `RequestProject/PictureSvg.lean` state the consistency of the rendering with the mathematics: sides drawn alike are exactly the sides that are glued. The TikZ figures in the present paper are redrawings of those, for typesetting reasons, and carry no independent authority.

8.4 What is not claimed

For the avoidance of doubt:

- only topological statements are formalised: no smooth, symplectic or complex structure appears anywhere;

- the cap is not claimed to be a manifold at its central fibre (it is one away from it, by Theorem 4.3(xiii));
- the squashing c is an input, not a derived degeneration; its canonicity among quotient homomorphisms of tori is stated as a remark and is not formalised;
- the identifications $W_\infty \cong M^5 \times S^1$ and $\pi_1(W_\infty) \cong \Gamma$ are noted as standard consequences and are not formalised;
- the fundamental group statement of Theorem 7.1 is formalised as an algebraic identity in the semidirect product $\mathbb{Z}^4 \rtimes_{T_0} \mathbb{Z}$ — the model of $\pi_1(M^5)$ — and not as a computation of the topological fundamental group.

A The statement of Theorem B in the machine

Below is PaperB of RequestProject/Paper.lean, transliterated into ASCII for typesetting (the file itself uses the usual Unicode notation). T4Collapse is the collapse of Construction 4.4, capRadius is ρ , cuspCapRim is ι_1 , capSlice is ι_r , cuspCapCentre is ι_0 , capScale is H and capPos is $\{\rho > 0\}$.

```
def PaperB : Prop :=
  CompactSpace CuspCapT4 /\ ConnectedSpace CuspCapT4 /\
  Continuous (capRadius T4Collapse) /\
  (forall p, capRadius T4Collapse p in Icc 0 1) /\
  (forall r in Icc 0 1, exists p, capRadius T4Collapse p = r) /\
  (IsClosedEmbedding cuspCapRim /\
   range cuspCapRim = {p | capRadius T4Collapse p = 1}) /\
  (forall r (hr : 0 < r), IsClosedEmbedding (capSlice T4Collapse r hr) /\
   range (capSlice T4Collapse r hr) = {p | capRadius T4Collapse p = r}) /\
  (IsClosedEmbedding cuspCapCentre /\
   range cuspCapCentre = {p | capRadius T4Collapse p = 0}) /\
  (Continuous (fun p => capScale T4Collapse p.1 p.2)) /\
  (forall p, capScale T4Collapse 1 p = p) /\
  (forall r t x, capScale T4Collapse 0 (capMk T4Collapse r t x)
   = cuspCapCentre (squash x)) /\
  (forall s q, capScale T4Collapse s (cuspCapCentre q) = cuspCapCentre q) /\
  Nonempty ((FiveManifold x Ioc 0 1) homeo (capPos T4Collapse))
```

B Where things are

file	contents
RequestProject/S6/LatticeMonodromy.lean	the matrices T_1, T_2, T_0, N
RequestProject/S6/MappingTorus.lean	mapping tori and $\mathbb{Z}^4 \rtimes_A \mathbb{Z}$
RequestProject/S6/HexagonOppositeQuotient.lean	double curves and pinch points
RequestProject/TorusMonodromy.lean	φ, M^5 , Theorem 3.2
RequestProject/CuspCap.lean	Construction 4.1, Theorem 4.3(i)–(xii)
RequestProject/CuspCapT4.lean	the cap of the question
RequestProject/CapPunctured.lean	the collar, Theorem 4.3(xiii)
RequestProject/PictureHexagon.lean	Theorem 5.1
RequestProject/Crystal/Lattice.lean	Γ , the cell, the charts, the refutations
RequestProject/Crystal/Prism.lean	the combinatorics of the drawn solid
RequestProject/AttachedProjectLink.lean	Theorem 7.1
RequestProject/Paper.lean	PaperA–PaperE, paper_certificate